

Assessing Logical Puzzle Solving in Large Language Models: Insights from a Minesweeper Case Study

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Abstract

Large Language Models (LLMs) have shown remarkable proficiency in language understanding and have been successfully applied to a variety of real-world tasks through task-specific fine-tuning or prompt engineering. Despite these advancements, it remains an open question whether LLMs are fundamentally capable of reasoning and planning, or if they primarily rely on recalling and synthesizing information from their training data. In our research, we introduce a novel task—Minesweeper—specifically designed in a format unfamiliar to LLMs and absent from their training datasets. This task challenges LLMs to identify the locations of mines based on numerical clues provided by adjacent opened cells. Successfully completing this task requires an understanding of each cell’s state, discerning spatial relationships between the clues and mines, and strategizing actions based on logical deductions drawn from the arrangement of the cells. Our experiments, including trials with the advanced GPT-4 model, indicate that while LLMs possess the foundational abilities required for this task, they struggle to integrate these into a coherent, multi-step logical reasoning process needed to solve Minesweeper. These findings highlight the need for further research to understand the nature of reasoning capabilities in LLMs under similar circumstances, and to explore pathways towards more sophisticated AI reasoning and planning models.

1 Introduction

Large Language Models (LLMs) have made remarkable strides in the Natural Language Processing (NLP) arena, capturing the spotlight with their multifaceted capabilities. These models have been effectively utilized across a spectrum of NLP tasks, including information extraction (Agrawal et al., 2022; Zhu et al., 2023b), answering arithmetic and common-sense questions (Li et al., 2022; Yuan et al., 2023; Imani et al., 2023), as well as aiding in

strategic planning (Yao et al., 2023b; Wang et al., 2023c) or acting as game agents (Callison-Burch et al., 2022; Wang et al., 2023a; Gupta, 2023). With growing scale, LLMs begin to exhibit “emergent abilities” (Wei et al., 2022a; Schaeffer et al., 2023), including marked improvements in their ability to follow instructions, perform multi-step reasoning and planning, and even extend to comprehending humor (OpenAI, 2023). Such flourishing capability has triggered excitement within the research community, cultivating the belief that LLMs could be instrumental in achieving Artificial General Intelligence (Bubeck et al., 2023; Zhang et al., 2023; Tu et al., 2023). Nevertheless, a crucial question remains underexplored: *To what extent can the reasoning capabilities of LLMs go beyond the scope of their training distribution?*

This question is critical as it uncovers if LLMs are reliable for sophisticated hands-off planning in various scenarios such as biomedical experimental design (Bran et al., 2023; O’Donoghue et al., 2023) and autonomous driving (Sha et al., 2023), where the agents have to deal with off-distribution inputs occasionally. Unfortunately, the answer is not straightforward, and the debate is going on (Huang et al., 2022; Helbling et al., 2023; Wu et al., 2023; Huang et al., 2023; Valmeekam et al., 2023; Stechly et al., 2023). The reason is that LLMs’ substantial parameter counts enable them to store and recall vast amounts of information from their training material (Gudibande et al., 2023; Rosenfeld et al., 2020; Kaplan et al., 2020; Brown et al., 2020), which may contain conventional reasoning datasets such as GSM8K (Cobbe et al., 2021), MultiArith (Roy and Roth, 2015), or StrategyQA (Geva et al., 2021). This could exaggerate LLMs’ reasoning abilities on such benchmarks (Qin et al., 2023; Deng et al., 2023), leading to false a promise when applying LLMs to practical scenarios.

Although attempts to develop new benchmarks that challenge LLMs suggest potential zero-shot

learning successes (Suzgun et al., 2023; Bao et al., 2023), there is an underlying concern that these benchmarks may not be distinct enough from the LLMs’ training data, thereby giving LLMs an undue advantage and masking their true abilities (Valmeekam et al., 2022). For instance, Gudibande et al. (2023) suggest that smaller LLMs only learn specific tasks when fine-tuned with data from larger models and that these improvements do not generalize well. Drawing an analogy between smaller and larger LLMs to humans implies that LLMs may not exhibit genuine intelligence or intrinsic reasoning beyond their training distribution (Liu et al., 2023; Bubeck et al., 2023). To robustly claim that LLMs possess reasoning abilities, they must be tested under a wider array of conditions. A fundamental approach to overcoming these challenges is to decouple the test inputs from the LLMs’ pre-training data, compelling them to rely on task descriptions for reasoning and planning rather than prior knowledge. This entails an environment with clear instructions, a rich but well-defined action space, and a distinct objective. Such a setup facilitates the verification of the reasoning process and qualitative performance assessment, while it also reduces the chances of successful random guessing.

Minesweeper, a well-known logic puzzle game illustrated in Figure 1, is proposed as an ideal testing ground under these considerations. The game presents a range of complexities: while its fundamental principles are straightforward and accessible to beginners, achieving proficiency requires advanced logical reasoning skills that go beyond mere pattern recognition on the game board. For LLMs, the challenge in Minesweeper lies in interpreting the states of individual cells, making use of numerical hints, and understanding the spatial interconnections between cells. This is essential for accurately determining the locations of mines and strategizing subsequent moves based on incomplete information—a process that is instinctive for human players. Despite LLMs’ prior knowledge of the universally known rules of Minesweeper, the transformation of game boards from visual representations to machine-readable text has been limited, with no existing format aligning with our representation. Consequently, LLMs must rely on their intrinsic logic for problem-solving by understanding and following the rules, as pattern matching from training data is impractical.

Building upon our initial motivation, we conduct a comprehensive set of experiments with LLMs in-

cluding GPT-3.5-16k, GPT-3.5-instruct, and GPT-4 (OpenAI, 2022, 2023), probing their inherent reasoning skills. We introduce the Minesweeper game in various formats to determine how different types of input influence LLM performance. Objective measures like the count and proportion of accurate moves, correctly flagged mines, and completely resolved boards were employed to gauge effectiveness. However, a critical aspect of our evaluation is the manual examination of the LLMs’ reasoning processes, inspecting the validity and logical soundness of their intermediate deductions. Our findings indicate that these models generally struggle to maintain consistent logical reasoning chains. The GPT-3.5 versions, in particular, tended to repeat information from provided examples or previous conversations, often failing to adjust to the updated board layouts. While GPT-4 showed improvements in response diversity, relevance, and coherence, it still faced issues with logical inconsistencies and incomplete reasoning. Our research categorizes errors in a detailed simulated environment, which sheds light on the specific skills required for solving logical puzzles, underscoring the importance of these competencies in future LLM-focused studies. Moreover, we notice that LLMs tend to generate ad-hoc sequences of actions rather than following logical reasoning chains based on action history, a finding not commonly reported, providing novel insights into LLM operational behaviors. To support forthcoming research and comprehensive evaluation of LLMs, we have published our code and data at <https://github.com/Yinghao-Li/Minesweeper-for-LLM>. We hope it contributes to LLMs’ broader understanding and future development.

2 Related Works

Recent studies have explored the reasoning capabilities of LLMs, with several investigations underway to understand (Qin et al., 2023; Bang et al., 2023; Liu et al., 2023; Xu et al., 2023; bench authors, 2023; Bao et al., 2023), enhance (Wei et al., 2022b; Imani et al., 2023), and leverage (Yao et al., 2023b; Sha et al., 2023) these abilities for various applications. Some critics argue that the datasets used for evaluation are overly simplistic and potentially compromised by inclusion in the training data of LLMs (OpenAI, 2023; Wei et al., 2023), casting doubt on the authenticity of these purported capabilities (Valmeekam et al., 2022; Bao et al., 2023). In response, new tasks and evaluation frameworks

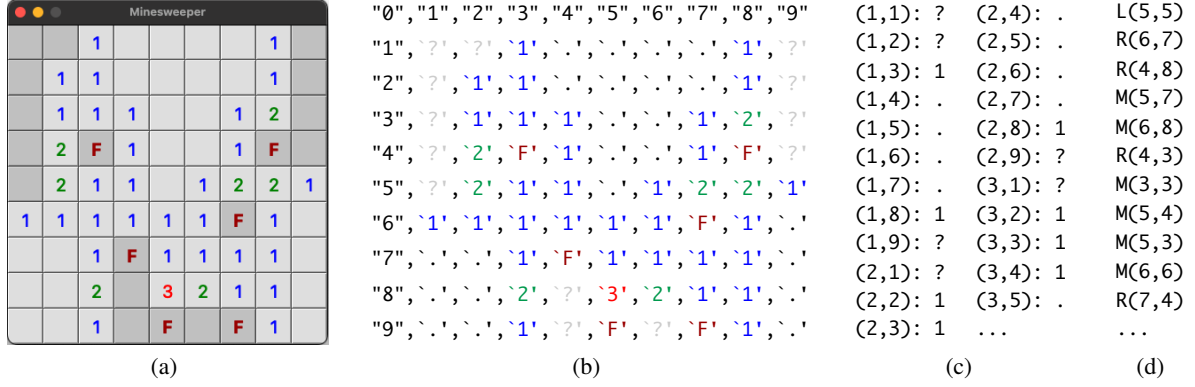


Figure 1: An example of Minesweeper on a 9×9 board containing 10 mines, along with its interaction format. Subfigure 1a displays the game’s GUI; Subfigure 1b shows a plain-text, table-formatted representation of the game board, enhanced with color for improved visualization; Subfigure 1c depicts the coordinate-based plain-text representation of the board; and Subfigure 1d provides a log of the player’s (in this case, the first author’s) actions, where “L”, “R”, and “M” denote left-click, right-click, and middle-click actions, respectively.

have been suggested (Valmeekam et al., 2022; Xu et al., 2023; Bao et al., 2023; Vervoort et al., 2023; Wang et al., 2023a). However, these tend to focus narrowly on specific aspects of arithmetic or commonsense reasoning, often presented in unstructured natural language that intersects with LLMs’ pre-existing world knowledge. This overlap presents a challenge in fully assessing LLMs’ inductive reasoning abilities. In contrast, the game of Minesweeper encompasses symbolic, arithmetic, and commonsense reasoning in a structured format lying outside the scope of LLMs’ training data, and thus demands genuine reasoning skills for successful resolution.

Along another track, the use of LLMs in gameplay has gained attention, with efforts to harness them for diverse games such as card games (Gupta, 2023; Guo et al., 2023), interactive narrative games (Callison-Burch et al., 2022; Tsai et al., 2023; Cui et al., 2023; Zhu et al., 2023a), chess (Noever et al., 2020; Stöckl, 2021; Suzgun et al., 2023; Feng et al., 2023), and video games like Minecraft (Wang et al., 2023a,b). Nonetheless, we contend that such games are intrinsically distinct from Minesweeper in both structure and the level of challenge they pose to LLMs. Card and interactive narrative games largely draw on common sense and narrative understanding, with a constrained range of actions. This limitation can obscure whether LLM decisions stem from deep reasoning or instinctual response. Chess, while offering a more extensive range of possibilities, presents an abstract playfield that typically requires tailored pre-training for LLMs to grasp its rules. Minecraft poses a further challenge with its

complex syntax and the need for meticulous, often biased, prompt engineering, making it an impressive display but not an accurate indicator of LLMs’ functional application. Minesweeper, by comparison, offers a straightforward yet strategically rich gameplay experience with clear objectives and direct evaluation criteria, making it an exemplary model for assessing the practical reasoning capabilities of LLMs.

3 Minesweeper

Minesweeper, a classic logic game, has been a part of the Windows operating system since around 1990s. Though the rule is simple, it features a rich and discrete action space, a clear goal, and direct feedbacks and evaluation metrics. The objective is straightforward: players must unveil all cells on a board peppered with hidden mines without detonating any. Each cell on the board can exist in one of four possible states (Figure 1b):

- **Unopened** (“?”): The state of these cells is unknown; they could transition into any of the other three states.
- **Numbered** (“1” to “8”): These cells reveal the count of adjacent mines, including diagonals, providing critical clues for safe navigation.
- **Blank** (“.”): These cells have no adjacent mines, often opening up larger areas of the board.
- **Flagged** (“F”): Cells suspected to contain mines are marked accordingly for safety.

Notably, the choice of symbols is arbitrary and should not impact the model performance, as they serve as mere representations of concepts.

During gameplay, players may execute one of three actions per turn: left-click (“L”) to reveal a cell’s contents, right-click (“R”) to flag a suspected mine, and middle-click (“M”) to verify the correctness of flagged cells based on adjacent numbers. On a beginner’s 9×9 board (Figure 1a), this results in 243 potential actions per round (Figure 1d), creating an extensive action landscape despite numerous actions being invalid (as listed in Table 3). The game provides immediate feedback for each action, with outcomes ranging from revealing or flagging cells to error messages, game endings or victories, allowing models to adjust their strategies in real-time. Victory is reached by correctly identifying and flagging all mines or revealing all non-mine cells, whereas failure occurs upon missteps such as erroneous left-click on mines or incorrect flag placements during middle-click verifications. With action space and game objective established, metrics such as the proportion of valid moves, the percentage of successfully solved games, and the average number of moves to resolve a board can be readily computed. In summary, Minesweeper serves as a less knowledge-intensive, more symbol-comprehension and basic math-reasoning benchmark, pivoting on spatial reasoning over pure information retrieval.

We employ two plain-text formats to present the game board when interacting with LLMs, aiming to understand LLMs’ capabilities of comprehending each of them and to reduce the potential influence of input format on the reasoning process. The first format is **table representation**, depicted in Figure 1b. Here, the game board is portrayed as a table with the states of each cell enclosed in $\LaTeX$ -style quotation marks (grave accent “`” and apostrophe “’”) to differentiate them from the cell indices and separators. Rows and columns are delineated by line break “\n” and comma “,” respectively, with indices for the first row and column serving as the coordinates for the cells. This format is intuitive for human interpretation; LLMs equipped with capabilities to understand tables, as suggested by the literature (Chen, 2023; Singha et al., 2023), should process it similarly. For comparison, we also employ **coordinate representation** (Figure 1c). It explicitly associates each cell’s coordinates with its state in the form of a look-up table, potentially offering a more direct description for LLMs to comprehend the board’s configuration.

The following sections first explore two critical skills required for Minesweeper agents (§ 4). Then,

Task	Representation	3.5-16k	3.5-instr
Board Navigation	table	66.7	53.3
	– ids	63.7	40.7
	+ example	64.3	47.0
	+ verification	66.3	-
	coordinate + example	82.3 89.7	66.7 61.0
Neighbor Counting	table	33.3	16.7
	coordinate	37.3	30.3

Table 1: Comparative analysis of GPT-3.5 variants in board understanding tasks. The performance are quantified using exact-matching accuracy percentages. The notation “– ids” indicates the omission of indices in the table representation, while “+ example” denotes the inclusion of additional examples in the prompts. The “+ verification” symbol refers to the application of the self-verification technique as described in (Weng et al., 2022). As GPT-3.5-instruct is not optimized for conversational contexts, the self-verification is not applicable.

they evaluate the proficiency of LLMs in playing Minesweeper, detailing objective scores and case studies that highlight their reasoning and planning capabilities (§ 5). Each section begins with a brief introduction of the experimental setup, with comprehensive details available in appendix A.

4 Board Understanding

4.1 Experiment Setup

A fundamental skill for LLMs in Minesweeper is the ability to comprehend the game board. To assess this capability, we conducted two types of straightforward experiments: **board navigation** and **neighbor counting**. In board navigation, LLMs are presented with a game board and a specific coordinate, and they are tasked to identify the corresponding cell state. For neighbor counting, LLMs are given an additional arbitrary state from the set $\{ '?', \cdot, '1', '2', \}$ and are required to calculate the number of occurrences of this state (ranging from 0 to 8) surrounding the provided coordinate. We annotated 100 randomly generated 9×9 boards, each containing 10 mines, with complete action steps (refer to Figure 1d). The boards were randomized at different stages of the game, either fully or partially revealed, and 3 sets of coordinates were selected randomly for the experiments.

4.2 Results

The performance of GPT-3.5-16k and GPT-3.5-instruct was evaluated using both table and coordinate representations, as shown in Table 1. With the

	Human	3.5-16k (NC)		3.5-16k (CH)		3.5-instr (CH)		4 (NC)	
	GUI	Table	Coord	Table	Coord	Table	Coord	Table	Coord
Game Outcome									
# Total Games	100							10	
• % Solved ($\uparrow$)	89.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
• % Failed ($\downarrow$)	11.0	17.0	35.0	17.0	35.0	19.0	41.0	70.0	70.0
Mine Identification									
# Total Mines	400							40	
• % Flagged ($\uparrow$)	93.0	2.8	11.7	8.0	9.5	11.7	23.3	30.0	45.0
Action Accuracy									
# Total Actions	514	825	679	700	687	813	642	41	52
• % Valid ($\uparrow$)	99.6	7.2	22.4	15.7	22.6	24.1	64.6	82.9	82.7
• % Repeated ($\downarrow$)	5.1	43.6	44.6	42.3	45.3	39.4	19.9	0.0	0.0
Reasoning Soundness									
# Selected Reasoning Chains	-	34	38	45	39	45	45	25	38
• % Valid Actions ($\uparrow$)	-	61.8	81.1	46.7	59.0	80.0	95.6	88.0	84.2
• % Accurate and Coherent Logic ($\uparrow$)	-	0.0	0.0	0.0	0.0	0.0	0.0	12.0	26.4

Table 2: The comparison of GPT model variants, prompting modes, and board representation formats on 5×5 boards with 4 mines. “#” indicates the number of instances; “%” represents the percentage ratio. The initial action, L(3, 3), is consistently excluded from the count. GPT-3.5-instruct is prompted using the compact history mode, as it is not optimized for chat-based interactions. The terms “Table” and “Coord” in the table refer to the table-format and coordinate-format representations of the Minesweeper board, respectively. The percentages of solved and failed cases do not sum to 100% due to instances where the game was not completed within 10 steps. The repetition is calculated independently for each board.

chains. A “valid” action is one that advances gameplay without activating a mine. Table 3 details a broad spectrum of negative examples. The metric for correctly flagged mines is based on the quantity of mines identified by the model that align with the actual mine locations at the game’s conclusion. A game is deemed “solved” when all mines are accurately marked and no extraneous cells are flagged. To assess the reasoning chains, we initially select 5 game boards where each model executed the highest number of valid actions. We then perform a manual inspection to ensure these chains are logically sound and directly pertinent to the gameplay. Due to the variable action counts across different models and boards, the totals of actions and reasoning chains differ, as Table 2 illustrates.

5.3 Results and Discussion

Table 2 showcases the performance comparisons between various GPT family models and a human benchmark, focusing on valid actions, correctly identified mines, completed games, and logical reasoning chains. In line with the findings on board comprehension, models utilizing coordinate representation exhibit superior performance across all

metrics compared to table representation. This supports our previous assertion that LLMs struggle with tables containing symbols not found in their training data, preferring familiar in-domain terminology as evidenced in the studies by Chen (2023) and Singha et al. (2023).

Objective Scores In assessing specific gameplay metrics, we first focus on three main areas: Game Outcomes, Mine Identification, and Action Accuracy. Among the GPT-3.5 variants, 3.5-instruct demonstrates better overall performance compared to 3.5-16k in these metrics, albeit with a marginally increased failure rate, which diverges from the board understanding results. A detailed analysis of GPT-3.5-instruct’s *initial* move reveals a consistent anomaly: the model persistently selects the R(2, 2) action, contrary to the given instruction of L(3, 3) (refer to Table 3 for invalid actions and Listing 5, Line 83 for the specific prompt). After receiving feedback, it switches to the L(1, 1) action. This action often uncovers fewer cells in our specially selected boards, usually just the chosen cell itself (as illustrated in Figure 3), which increases the probability of the next action being valid. This be-



With the current state of the board, there are several safe moves that can be made based on the information provided by the numbered cells. The cells along the edge that are showing a '1' next to a blank cell can be safely assumed to have a mine in the adjacent unopened cell. Specifically, the cell at (1,3) has only one adjacent unopened cell at (1,2), which must contain a mine. To progress in the game without risking opening a mine, we can flag the cell at (1,2) as it is the only unopened adjacent cell to (1,3).

The cell at (4,1) is showing a '1', indicating there is one mine adjacent to it. Since the only unopened adjacent cell is at (5,1), we can deduce that the mine must be in that location. Therefore, we can safely flag cell (5,1) with a right-click.

Observing the board, the cell at (2,1) shows a '2', indicating there are two mines adjacent to it. One mine is already flagged at (1,2). The only other unopened cell next to it is at (1,1), so we can conclude that the other mine must be at (1,1). Thus, the safe action is to flag cell (1,1).

The cell at (4,4) shows a '1', which means there is one mine adjacent to it. Since all surrounding cells are revealed except for the one at (5,4), we can deduce that (5,4) must contain the mine. We should therefore flag cell (5,4) with a right-click.

Looking at cell (2,1), we can see it has a '2' which indicates there are two mines in the neighboring cells. Since one mine is already flagged at (1,1), the other mine must be in one of the unopened cells adjacent to (2,1). Cell (2,2) also has a '2', and given that there's already a mine flagged at (1,2), the second mine adjacent to (2,2) can only be at (3,1) because (3,2) is opened and cannot contain a mine. Thus, we can safely flag the cell at (3,1) as containing a mine.

The corrected action based on the previous reasoning should be to flag the cell at (3,1) as it is suspected to contain a mine. This is deduced from the fact that the '2' in both cells (2,1) and (2,2) must be satisfied by having two mines adjacent to them, and with the given flags and opened cells, the only possible location for the second mine around both of these cells is at (3,1).

Figure 5: This figure presents a case study of the reasoning sequences formulated by GPT-4 with coordinate board representation during action planning. The GUI on the left shows the board states and actions taken by the agent during the game. Elements highlighted in blue represent accurate facts and logical inferences as assessed by human evaluation, while those marked in red indicate incorrect observations or illogical conclusions. Notably, the final generated action “F(3,1)” deviates from the permissible action formats, resulting in the termination of the game.

Reasoning Chains The foregoing discussion highlights that merely counting valid actions is not an adequate measure of the models’ reasoning skills. Therefore, we shift our focus to the logical coherence of their reasoning chains. A review of the final section in Table 2 reveals that the GPT-3.5 variants struggle to generate coherent reasoning. For instance, in a typical case from GPT-3.5-16k, shown in Figure 4, the model correctly identifies cell ‘4’ at (2,4) and understands that there are four mines around, but fails to grasp the concept of “neighboring cells” or to accurately count the surrounding cell states, as indicated by the statement marked in red. Furthermore, the statement in golden font is not well-grounded, making it challenging to discern any logical connection between its first and second halves. This sentence appears more like an unsuccessful attempt to replicate our in-context example “Given that there’s just one adjacent mine, it’s logical to deduce that the unopened cell at (2,3) contains the mine” (refer to

Listing 5, Line 51) rather than a display of genuine reasoning. While this example is specifically for GPT-3.5-16k, similar errors are prevalent across all GPT-3.5 outputs. It appears that despite possessing a certain degree of individual capabilities in elementary symbolic understanding, geographical reasoning, and arithmetic calculation, as suggested by the board understanding results, the GPT-3.5 models fall short in integrating these skills cohesively for action planning, resulting in a fragmented reasoning chain.

GPT-4 exhibits more encouraging results compared to its predecessors. While its achievements are modest, GPT-4 demonstrates a certain level of logical understanding in this unfamiliar task, suggesting an ability to grasp the game’s rules and dynamics beyond mere replication. To explore the extent and origin of this reasoning, we delve deeper into GPT-4’s reasoning chains, analyzing a representative case in Figure 5. It shows that GPT-4 can logically reason if the number of neighboring un-

opened cells equals the remaining unflagged mines near a numbered cell, specifically in simple scenarios like ``1'` and ``2'`, and particularly around the board's corners (e.g., $(1, 2)$ or $(5, 4)$). Yet, even in such basic situations, GPT-4 shows around 50% fails. The model struggles particularly with configurations where exactly three unopened cells surround a ``3'` cell and other complex setups, as seen in the lower rows of the example. To determine if GPT-4's ability stems from the rules or the examples, we conducted a control experiment by omitting Example 1 from the initial prompt (Listing 5, Line 43), which closely mirrors the successful scenarios and is the sole example of left-clicking. This led to a dramatic 90% reduction in coherent reasoning with the coordinate board representation, indicating the significant role of examples in GPT-4's reasoning process, likely by providing a familiar context for reference and imitation.

Another notable point from Figure 5 is GPT-4's non-linear action sequence. For a human player, following the initial action of $R(1, 2)$, the logical next step would be to mark the adjacent unopened cell as a mine, i.e., $R(1, 1)$. However, GPT-4 instead opts for $R(5, 1)$, an irrational choice that strays from the focus of the previous action, before returning to $(1, 1)$. This behavior suggests that GPT-4 might not fully consider its historical actions and lacks the capability for long-term, multi-step planning, a crucial aspect in evaluating an agent's intrinsic reasoning abilities as opposed to mere summarization capabilities.

It is important to note that the results discussed were derived from experiments conducted on 5×5 boards, and it is reasonable to anticipate that performance may further deteriorate on larger boards with more flexible mine arrangements.

6 Conclusion

In this study, we assessed the capabilities of LLMs in solving logical puzzles by examining their performance in the game Minesweeper, a task for which they have received limited exposure during the pre-training phase. This approach aims to assess the intrinsic reasoning capabilities of LLMs. Our experiments reveal that GPT-3.5 models exhibit basic abilities in spatial navigation, symbol recognition, and counting. However, they struggle with multi-step planning and generating coherent reasoning chains. GPT-4 shows improvement in simpler scenarios with shorter logic chains but still

encounters issues with hallucinations and inconsistent context awareness. We find that GPT-4 can learn from examples and apply this knowledge to similar situations but is less adept at deducing underlying rules and applying them to novel scenarios, unlike GPT-3.5 models which tend to replicate examples. Our conclusion seems to be in correlation with [Yadlowsky et al. \(2023\)](#): LLMs do not produce new knowledge (a situation where logical reasoning is necessary); they are just (good) handlers of existing experience.

We do not intend to diminish the significance of LLMs or their transformative impact on natural language processing. Rather, we believe it is premature to consider LLMs as an intelligent object and a threat to human society. They resemble a comprehensive dictionary, useful for reference but lacking in comprehension. We hope that our work can inspire future research on this topic, and we will continue exploring the capabilities of LLMs.

Limitations

In our study, we employed two board representation methods: table and coordinate representations. Our results indicate that coordinate representation is more effective for LLM comprehension and offers a wider range for evaluating the model's reasoning abilities. However, we did not explore other board representation methods, such as different table formats (e.g., HTML, Markdown) or image-based approaches. According to [Singha et al. \(2023\)](#), alternative table formats may not significantly differ in effectiveness from our chosen method. Still, it's possible that other representations could be more suitable for our specific context. Moreover, our initial attempts to use an image-based Web interface with board GUI were not successful. Therefore, we focus on text-based representations in this study. While advanced prompting methods like Program of Thoughts (PoT, [Chen et al., 2022](#)) or Tree of Thoughts (ToT, [Yao et al., 2023a](#)) may boost performance, their use could divert from assessing the true reasoning abilities of LLMs and their applicability in simplifying complex issues without extensive external input. Hence, we deferred the application of these sophisticated techniques to future studies.

Another area for exploration is fine-tuning LLMs like Llama-2-70b ([Touvron et al., 2023](#)) with specific reasoning capabilities on our dataset labeled with human actions. The idea is that if a fine-tuned

model outperforms the original or GPT-4 when the fine-tuning and testing data match in terms of symbol usage, board size, and representation, and does not improve with variations in these factors, we can infer that the “reasoning capability” is tied to the training data and not broadly generalizable. However, due to constraints in computational resources, we haven’t conducted these experiments, leaving them for future research.

At last, we would like to emphasize that we can easily extend the Minesweeper dataset to a scale of $\binom{n_{\text{row}} \times n_{\text{column}}}{n_{\text{mine}}}$ with zero effort. Nonetheless, we keep a relatively small scale of test cases as it is sufficient to support our conclusion. With the future development of LLMs, a larger evaluation set might be desired for a more robust and unbiased objective metric-based evaluation.

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A Experiment Setup

A.1 GPT Versions

In this paper, we discuss three distinct versions of the GPT model: GPT-3.5-16k, GPT-3.5-instruct, and GPT-4. The term GPT-3.5-16k is used to denote the “gpt-3.5-turbo-16k” model as listed on the OpenAI API model webpage, specifically the checkpointed 0613.¹ The GPT-3.5-instruct model refers to the “gpt-3.5-turbo-instruct” variant, with the checkpoint version dated 2023-09-15. Lastly, GPT-4 represents the “gpt-4” model, utilizing also the 0613 checkpoint version. All mentioned checkpoints are hosted on Microsoft Azure.² The model temperature is set to 0 in all cases.

A.2 Board Understanding

In our board understanding experiments, we initially generate 1,000 randomly arranged boards, each featuring a 9×9 grid with 10 hidden mines. Notably, the central cells, positioned at (5,5), are always mine-free. From this pool, we select 100 boards, ensuring that the initial click at the center L(5,5) reveals at least 10 cells. Subsequently, we conduct gameplay using the graphical user interface depicted in Figure 1a. The actions of the annotators, who are average players in Minesweeper and unaware of the board’s layout beyond the initial instruction to click the center, are recorded as shown in Figure 1d. Although not all trials are successful—with about 90% boards being solved and the remainder failing due to human errors or 50-50 guesses—the entire action history, including unsuccessful attempts, is preserved for analysis.

For each board, we randomly choose an action (excluding the first and last from the history) and associate board state as the focus. We then select 3 random coordinates per board, setting two challenges for LLMs: to determine the status of each selected cell (board navigation), and to count the number of specific symbols within $\{\text{'?'}, \text{'.'}, \text{'1'}, \text{'2'}\}$ in their neighbors, including diagonally adjacent ones (neighbor counting). This process yields 300 test cases for each task.

As outlined in § 4, we employ two distinct board representations: table and coordinate. Each required a slightly different approach in prompting. Specifically, the table representation uses $\text{\LaTeX}$ -style quotation marks, while the coordinate representation does not use quotations for board cell

states and employs regular double quotes in the necessary positions in the game description. Examples illustrating these differences are provided below.

Listing 1: An example of board navigation prompting with table representation.

```
1 You will be presented with a 9 by 9 board for the
  Minesweeper game. The board is wrapped in a 10
  by 10 table, where the first row and the first
  column with numbers in double quotation marks
  are the row and column indices. A coordinate (x
  ,y) represents the cell at the x-th row and y-
  th column, where x and y, starting from 1, are
  the row and column indices, respectively. The
  state of each cell is represented by the
  following symbols:
2 - '.' represents a blank cell.
3 - '1' to '8' represents numbered cells with that
  number of mines in the adjacent cells.
4 - 'F' represents a flagged cell.
5 - '?' represents an unopened cell.
6
7 --- EXAMPLES ---
8 --- PARTIAL BOARD ---
9 "0","1","2","3","4"
10 "1","?","?","1","."
11 "2","?","?","3","1"
12 "3","?","F","2","F"
13 "4","?","2","2","1"
14
15 QUESTION: What is the cell at coordinate (1,3)?
16 ANSWER: '1'
17
18 --- PARTIAL BOARD ---
19 "0","1","2","3","4"
20 "1","?","?","1","."
21 "2","?","?","3","1"
22 "3","?","F","2","F"
23 "4","?","2","2","1"
24
25 QUESTION: What is the cell at coordinate (4,1)?
26 ANSWER: '?'
27
28 --- END OF EXAMPLES ---
29
30 --- CURRENT BOARD ---
31 "0","1","2","3","4","5","6","7","8","9"
32 "1","?","?","?","?","1",".", ".", ".", "."
33 "2","?","?","?","?","1",".", ".", ".", "."
34 "3","?","1","1","1","1",".", ".", ".", "."
35 "4","?","1",".", ".", ".", ".", ".", ".", "."
36 "5","?","2","1","1",".", ".", ".", ".", "."
37 "6","?","?","?","1","1","2","3","2","1"
38 "7","?","?","?","?","?","?","?","?","?"
39 "8","?","?","?","?","?","?","?","?","?"
40 "9","?","?","?","?","?","?","?","?","?"
41
42 QUESTION: What is the cell at coordinate (2,1)?
43 ANSWER:
```

Listing 2: An example of board navigation prompting with coordinate representation.

```
1 You will be presented with a 9 by 9 board for the
  Minesweeper game, which is divided into cells.
  The cells are presented as "coordinate: state"
  mappings. A coordinate (x,y) represents the
  element at the x-th row and y-th column, where
  x and y, starting from 1, are the row and
  column indices, respectively. The state of each
  cell is represented by the following symbols:
2 - "." represents a blank cell.
3 - "1" to "8" represents numbered cells with that
  number of mines in the adjacent cells.
4 - "F" represents a flagged cell.
5 - "?" represents an unopened cell.
6
7 --- EXAMPLES ---
8 --- PARTIAL BOARD ---
9 (1,1): ?
10 (1,2): ?
11 (1,3): ?
```

¹<https://platform.openai.com/docs/models>

² *.openai.azure.com

```

12 (1,4): ?
13 (2,1): ?
14 (2,2): ?
15 (2,3): ?
16 (2,4): ?
17 (3,1): ?
18 (3,2): ?
19 (3,3): ?
20 (3,4): 1
21 (4,1): ?
22 (4,2): ?
23 (4,3): ?
24 (4,4): 1
25
26 QUESTION: What is the cell at coordinate (3,4)?
27 ANSWER: "1"
28
29 --- END OF EXAMPLES ---
30
31 --- CURRENT BOARD ---
32 (1,1): ?
33 (1,2): ?
34 (1,3): ?
35 (1,4): ?
36 (1,5): 1
37 (1,6): .
38 (1,7): .
39 (1,8): .
40 (1,9): .
41 (2,1): ?
42 (2,2): ?
43 (2,3): ?
44 (2,4): ?
45 (2,5): 1
46 (2,6): .
47 (2,7): 1
48 (2,8): 1
49 (2,9): 1
50 (3,1): ?
51 (3,2): 1
52 (3,3): 1
53 (3,4): 1
54 (3,5): 1
55 (3,6): .
56 (3,7): 1
57 (3,8): F
58 (3,9): 1
59 (4,1): ?
60 (4,2): 1
61 (4,3): .
62 (4,4): .
63 (4,5): .
64 (4,6): .
65 (4,7): 1
66 (4,8): 1
67 (4,9): 1
68 (5,1): ?
69 (5,2): 2
70 (5,3): 1
71 (5,4): 1
72 (5,5): .
73 (5,6): .
74 (5,7): .
75 (5,8): .
76 (5,9): .
77 (6,1): ?
78 (6,2): ?
79 (6,3): ?
80 (6,4): 1
81 (6,5): 1
82 (6,6): 2
83 (6,7): 3
84 (6,8): 2
85 (6,9): 1
86 (7,1): ?
87 (7,2): ?
88 (7,3): ?
89 (7,4): ?
90 (7,5): ?
91 (7,6): ?
92 (7,7): ?
93 (7,8): ?
94 (7,9): ?
95 (8,1): ?
96 (8,2): ?
97 (8,3): ?
98 (8,4): ?

```

```

99 (8,5): ?
100 (8,6): ?
101 (8,7): ?
102 (8,8): ?
103 (8,9): ?
104 (9,1): ?
105 (9,2): ?
106 (9,3): ?
107 (9,4): ?
108 (9,5): ?
109 (9,6): ?
110 (9,7): ?
111 (9,8): ?
112 (9,9): ?
113
114 QUESTION: What is the cell at coordinate (2,1)?
115 ANSWER:

```

Listing 3: An example of neighbor counting prompting with table representation.

```

1 You will be presented with a 9 by 9 board for the
  Minesweeper game. The board is wrapped in a 10
  by 10 table, where the first row and the first
  column with numbers in double quotation marks
  are the row and column indices. A coordinate (x
  ,y) represents the cell at the x-th row and y-
  th column, where x and y, starting from 1, are
  the row and column indices, respectively. The
  state of each cell is represented by the
  following symbols:
2 - '.' represents a blank cell.
3 - `1' to `8' represents numbered cells with that
  number of mines in the adjacent cells.
4 - `F' represents a flagged cell.
5 - `?' represents an unopened cell.
6
7 --- EXAMPLES ---
8 --- PARTIAL BOARD ---
9 "0", "1", "2", "3", "4", "5"
10 "1", `?', `?', `?', `?', `?'
11 "2", `?', `?', `?', `?', `?'
12 "3", `?', `?', `?', `1', `1'
13 "4", `?', `?', `?', `1', `.'
14 "5", `?', `?', `?', `1', `.'
15
16 QUESTION: How many cells `F' are neighbors (
  including diagonal) of the cell with coordinate
  (2,1)?
17 ANSWER:
18
19 To find out how many cells with the value `1' are
  neighbors of the cell with coordinate (2,1), we
  need to look at the 8 neighboring cells of
  (2,1). These coordinates are:
20 (1,1), (1,2), (3,1), (3,2), (2,2).
21
22 Now, we will check the values of these cells on the
  given Minesweeper board:
23
24 (1,1) = `?'
25 (1,2) = `?'
26 (3,1) = `?'
27 (3,2) = `?'
28 (2,2) = `?'
29
30 All of these neighboring cells are `?'. So, the
  number of cells with `1' that are neighbors of
  the cell (2,1) is:
31
32 ANSWER: 0.
33
34 --- END OF EXAMPLES ---
35
36 --- CURRENT BOARD ---
37 "0" "1" "2" "3" "4" "5" "6" "7" "8" "9"
38 "1", `?', `?', `?', `?', `1', `.', `.', `.', `.'
39 "2", `?', `?', `?', `?', `1', `.', `1', `1', `1'
40 "3", `?', `1', `1', `1', `1', `.', `1', `F', `1'
41 "4", `?', `1', `.', `.', `.', `.', `1', `1', `1'
42 "5", `?', `2', `1', `1', `.', `.', `.', `.', `.'
43 "6", `?', `?', `?', `1', `1', `2', `3', `2', `1'
44 "7", `?', `?', `?', `?', `?', `?', `?', `?', `?'
45 "8", `?', `?', `?', `?', `?', `?', `?', `?', `?'
46 "9", `?', `?', `?', `?', `?', `?', `?', `?', `?'
47

```

```

48 QUESTION: How many cells '1' are neighbors (
    including diagonal) of the cell with coordinate
    (2,1)?
49 Let's think step by step.
50 ANSWER:

```

Listing 4: An example of neighbor counting prompting with coordinate representation.

```

1 You will be presented with a 9 by 9 board for the
  Minesweeper game, which is divided into cells.
  The cells are presented as "coordinate: state"
  mappings. A coordinate (x,y) represents the
  element at the x-th row and y-th column, where
  x and y, starting from 1, are the row and
  column indices, respectively. The state of each
  cell is represented by the following symbols:
2 - "." represents a blank cell.
3 - "1" to "8" represents numbered cells with that
  number of mines in the adjacent cells.
4 - "F" represents a flagged cell.
5 - "?" represents an unopened cell.
6
7 --- EXAMPLES ---
8 --- PARTIAL BOARD ---
9 (1,1): F
10 (1,2): ?
11 (1,3): ?
12 (1,4): F
13 (2,1): ?
14 (2,2): ?
15 (2,3): ?
16 (2,4): ?
17 (3,1): ?
18 (3,2): F
19 (3,3): ?
20 (3,4): 1
21 (4,1): ?
22 (4,2): ?
23 (4,3): ?
24 (4,4): 1
25
26 QUESTION: How many cells "F" are neighboring (
    including diagonally) the cell with coordinates
    (2,1)?
27 Let's think step by step
28 ANSWER:
29
30 To find out how many cells with the value "1" are
  neighbors of the cell with coordinate (2,1), we
  need to look at the 8 neighboring cells of
  (2,1). These coordinates are:
31 (1,1), (1,2), (3,1), (3,2), (2,2).
32
33 Now, we will check the values of these cells on the
  given Minesweeper board:
34
35 (1,1) = F
36 (1,2) = ?
37 (3,1) = ?
38 (3,2) = F
39 (2,2) = ?
40
41 From of these neighboring cells (1,1) and (3,2) are
  "F". So, the number of cells with "1" that are
  neighbors of the cell (2,1) is:
42
43 ANSWER: 2.
44
45 --- END OF EXAMPLES ---
46
47 --- CURRENT BOARD ---
48 (1,1): ?
49 (1,2): ?
50 (1,3): ?
51 (1,4): ?
52 (1,5): 1
53 (1,6): .
54 (1,7): .
55 (1,8): .
56 (1,9): .
57 (2,1): ?
58 (2,2): ?
59 (2,3): ?
60 (2,4): ?
61 (2,5): 1

```

```

62 (2,6): .
63 (2,7): 1
64 (2,8): 1
65 (2,9): 1
66 (3,1): ?
67 (3,2): 1
68 (3,3): 1
69 (3,4): 1
70 (3,5): 1
71 (3,6): .
72 (3,7): 1
73 (3,8): F
74 (3,9): 1
75 (4,1): ?
76 (4,2): 1
77 (4,3): .
78 (4,4): .
79 (4,5): .
80 (4,6): .
81 (4,7): 1
82 (4,8): 1
83 (4,9): 1
84 (5,1): ?
85 (5,2): 2
86 (5,3): 1
87 (5,4): 1
88 (5,5): .
89 (5,6): .
90 (5,7): .
91 (5,8): .
92 (5,9): .
93 (6,1): ?
94 (6,2): ?
95 (6,3): ?
96 (6,4): 1
97 (6,5): 1
98 (6,6): 2
99 (6,7): 3
100 (6,8): 2
101 (6,9): 1
102 (7,1): ?
103 (7,2): ?
104 (7,3): ?
105 (7,4): ?
106 (7,5): ?
107 (7,6): ?
108 (7,7): ?
109 (7,8): ?
110 (7,9): ?
111 (8,1): ?
112 (8,2): ?
113 (8,3): ?
114 (8,4): ?
115 (8,5): ?
116 (8,6): ?
117 (8,7): ?
118 (8,8): ?
119 (8,9): ?
120 (9,1): ?
121 (9,2): ?
122 (9,3): ?
123 (9,4): ?
124 (9,5): ?
125 (9,6): ?
126 (9,7): ?
127 (9,8): ?
128 (9,9): ?
129
130 QUESTION: How many cells "1" are neighbors (
    including diagonal) of the cell with coordinate
    (2,1)?
131 Let's think step by step.
132 ANSWER:

```

In our analysis of the prompts, we employ the Chain-of-Thought (COT) technique, as described by [Wei et al. \(2022b\)](#), for the neighbor counting task but not for board navigation. The rationale behind this is that neighbor counting necessitates multi-step action: initially identifying the neighboring cells of a given target cell and subsequently tallying a specific symbol within these neighbors.

In contrast, the board navigation task is straightforward, involving a direct query about a specified target cell. We have observed that incorporating COT into the neighbor counting process notably enhances its performance.

Furthermore, we have made the inclusion of in-context examples optional for the board navigation task, but mandatory for neighbor counting. The need for examples in neighbor counting arises from the models' propensity to generate unclear outputs in this task. By providing an example, we aim to guide the models towards producing more interpretable responses.

A.3 Minesweeper

The detailed setup of the Minesweeper experiment is presented in § 5. We hereby provide the actual initial prompts with table-formatted board representation used in our experiments. The coordinate representation closely mirrors the variations found in Listing 3 and Listing 4 and will not be omitted.

Listing 5: Initial prompt for Minesweeper gameplay.

```
1 >> SYSTEM:
2 You are a helpful assistant who is good at playing
  Minesweeper.
3
4 >> USER:
5 --- MINESWEEPER INTRODUCTION ---
6 In Minesweeper, 4 hidden mines are scattered
  throughout a 5 by 5 board, which is divided
  into cells. The rows are separated by newlines,
  and columns by commas. The board is structured
  as a 6 by 6 table, with the first row and
  column labeled using numbers in double
  quotation marks to indicate row and column
  indices. Cells have multiple possible states:
7 - Unopened cells (represented by '?', which cover
  the board at the start of the game, can also be
  made by removing flags)
8 - Numbered cells (represented by '1' to '8', which
  indicate the number of mines in the eight
  neighboring cells, including those diagonally
  adjacent)
9 - Blank cells (represented by '.', which have no
  neighboring mines)
10 - Flagged cells (represented by 'F', which are
  marked by the player to indicate a potential
  mine location)
11
12 A player selects a cell to open it. If a player
  opens a cell containing a mine, the game ends
  in a loss. Otherwise, the opened cell displays
  either a number, indicating the number of mines
  diagonally and/or adjacent to it, or a blank
  tile (sometimes shown as a 0), and all adjacent
  cells will automatically be opened. To win a
  game of Minesweeper, all non-mine cells must be
  opened without opening a mine.
13
14 --- ACTION OPTIONS ---
15 There are three permissible actions in Minesweeper:
16
17 - Left-click an unopened cell ('?') to reveal it.
18 - Right-click an unopened cell ('?') to place a flag
  or a flagged cell ('F') to remove the flag.
19 - Middle-click on a numbered cell ('1' to '8') to
  unveil its neighboring cells, but only if all
  adjacent mines have been correctly flagged. If
  any flags are misplaced, you'll lose the game.
20
21 --- ACTION FORMAT ---
```

```
22 Each of your actions should be formatted as "A(row,
  col)", where:
23 - "A" represents the action type: "L" denotes a left
  -click, "R" indicates a right-click, and "M"
  signifies a middle-click.
24 - "row" specifies the row number of the targeted
  cell.
25 - "col" details the column number of the targeted
  cell.
26 For instance, an action like "L(1,2)" translates to
  a left-click on the cell located at the first
  row and second column.
27
28 please ensure:
29 - You do not duplicate actions.
30 - You submit only one action at a time.
31
32 --- CURRENT BOARD ---
33 ---
34 "0","1","2","3","4","5"
35 "1","?","?","?","?","?"
36 "2","?","?","?","?","?"
37 "3","?","?","?","?","?"
38 "4","?","?","?","?","?"
39 "5","?","?","?","?","?"
40 ---
41
42 --- EXAMPLES ---
43 Example 1:
44 --- PARTIAL BOARD ---
45 "0","1","2","3","4"
46 "1",".",",1","?","?"
47 "2",".",",1","?","?"
48 "3",".",",1","2","4"
49 "4",".",",.",",.",",2"
50
51 REASONING: The cell located at (3,2) displays the
  number '1', indicating there's a single mine in
  its neighboring cells. Among these neighbors,
  7 are opened as empty or numbered, and only one
  remains unopened at (2,3). Given that there's
  just one adjacent mine, it's logical to deduce
  that the unopened cell at (2,3) contains the
  mine. Thus, our next move is to flag the cell
  at (2,3) with a right-click.
52
53 ACTION: R(2,3)
54
55 Example 2:
56 --- PARTIAL BOARD ---
57 "0","4","5","6","7"
58 "1",".",",1","?","?"
59 "2",".",",1","F","?"
60 "3",".",",1","2","4"
61 "4",".",",.",",.",",2"
62
63 REASONING: The cell at (1,5) displays the number
  '1', indicating a single adjacent mine. Upon
  examining its neighboring cells, only the cell
  at (2,6) is flagged with 'F'. This suggests
  that all mines neighboring (1,5) have been
  identified. Consequently, we can safely unveil
  the other surrounding cells by middle-clicking
  on (1,5).
64
65 ACTION: M(1,5)
66
67 Example 3:
68 --- PARTIAL BOARD ---
69 "0","1","2","3","4"
70 "1","1","1","2","?"
71 "2","?","?","?","?"
72 "3","?","?","?","?"
73
74 REASONING: The cell at (1,1) indicates there's a
  single mine amongst its neighbors. Examining
  the cells adjacent to it, both (2,1) and (2,2)
  remain unopened, implying one of them contains
  a mine. Similarly, the cell at (1,2) displays a
  '1', suggesting that out of (2,1), (2,2), and
  (2,3), one holds a mine. Since one of (2,1) or
  (2,2) already contains a mine, it becomes
  evident that (2,3) is mine-free. We can then
  safely uncover (2,3) with a left-click.
75
76 ACTION: L(2,3)
77
```

Type of Invalid Actions	Example	Feedback
Coordinate		
Coordinate Out of Bound	"L(0,10)" on a 9×9 board	Invalid Coordinates! Please make sure your coordinate are within [1, 9] for rows and [1, 9] for columns.
Game Initialization		
Starting by Right-Clicking	"R(5,5)" on a new board	Please begin by left-clicking on the center cell.
Starting by Middle-Clicking	"M(5,5)" on a new board	Please begin by left-clicking on the center cell.
Left-Clicking		
Left-Clicking on Blank Cell	"L(5,5)" where (5,5) is opened as `.`	Invalid action: Cannot left-click a blank cell. Left-click is only for unopened cells (`?').
Left-Clicking on Flagged Cell	"L(5,5)" where (5,5) is flagged as `F'	Invalid action: Cannot left-click a flagged cell. Left-click is only for unopened cells (`?').
Left-Clicking on Numbered Cell	"L(5,5)" where (5,5) is opened as `2'	Invalid action: Cannot left-click a numbered cell. Left-click is only for unopened cells (`?').
Middle-Clicking		
Middle-Clicking on Blank Cell	"M(5,5)" where (5,5) is opened as `.`	Invalid action: Cannot middle-click a blank cell. Middle-click is only for numbered cells (`1' to `8').
Middle-Clicking on Flagged Cell	"M(5,5)" where (5,5) is flagged as `F'	Invalid action: Cannot middle-click a flagged cell. Middle-click is only for numbered cells (`1' to `8').
Middle-Clicking on Unopened Cell	"M(5,5)" where (5,5) is unopened (`?')	Invalid action: Cannot middle-click an unopened cell. Middle-click is only for numbered cells (`1' to `8').
Middle-Clicking When Numbered Cell Has no Flagged Neighbor		Error: No flagged cells detected nearby. Flag adjacent mines before middle-clicking.
Middle-Clicking When # Flagged Neighbor Mismatches Cell Number		Error: Flag count mismatch. Ensure all adjacent mines are flagged before middle-clicking.
Right-Clicking		
Right-Clicking on Blank Cell	"R(5,5)" where (5,5) is opened as `.`	Invalid action: Cannot right-click a blank cell. Right-click is only for unopened cells (`?') or flagged cells (`?').
Right-Clicking on Numbered Cell	"R(5,5)" where (5,5) is opened as `1'	Invalid action: Cannot right-click a numbered cell. Right-click is only for unopened cells (`?') or flagged cells (`?').

Table 3: Types and feedbacks for invalid actions.

```

78 --- END OF EXAMPLES ---
79
80 --- RESPONSE GUIDE ---
81 Let's think step by step.
82
83 Begin by detailing your rationale in the "REASONING"
   section, followed by specifying your move
   using the previously mentioned format in the "
   ACTION" section. Ensure your reasoning doesn't
   exceed 200 words. To commence the game, left-
   click the center cell at (3, 3).
84
85 REASONING:
86 ACTION:

```

In subsequent iterations, the prompts are modified based on the model’s responses. When the model generates a valid action, the updated game board is presented, followed by a request for the model to initiate the next move. In contrast, if the action is invalid, an error message is displayed, prompting the model to select an alternative action. Table 3 enumerates the various types of invalid ac-

tions and their corresponding feedback tailored for LLMs. Note that these actions do not directly violate the game rules. Rather, they are deemed invalid within the scope of our experiments because they are logically ungrounded and do not contribute to the progression of the game.

B Obfuscation and Numerical Expression

To study the impact of existing Minesweeper-related description within LLMs’ training materials, we conduct additional experiments to assess the impact of varying the game descriptions and the representation of numerical values on the performance of the model. In the obfuscation experiment, we replaced all instances of “Minesweeper” with descriptions of a generic logical puzzle, substituting “mine” with “thorn” and standard mouse-click

	Original Representation		Game Obfuscation		Numerical Expression	
	Table	Coord	Table	Coord	Table	Coord
# Total Games	100					
% Solved	0	0	0	0	0	0
% Failed	17	35	13	47	15	41
# Total Mines	400					
% Flagged Mines	2.8	11.7	4.5	13.8	7.0	14.5
# Total Actions	825	679	833	653	790	672
% Valid	7.2	22.4	7.3	28.8	9.9	28.6
% Repeated	43.6	44.6	57.5	44.1	44.6	39.7

Table 4: Ablation study on game obfuscation and numerical expressions. The results are all achieved using GPT-3.5-16k with NC mode.

actions within game description and action feedbacks with abstract symbols (e.g., “L/R/M”). In the other set of experiments, Arabic numerals that indicate the numbers of surrounding mines (e.g., 1,2,...,8) were converted to their Roman counterparts (e.g., I, II, ..., VIII), with other elements remaining unchanged. The outcomes are summarized in Table 4. They show that these variations have minimal effect on model performance, indicating that these factors do not significantly hinder the model’s reasoning abilities. Our findings suggest that the general exposure to Minesweeper game rules across the internet does not provide a distinct advantage for GPT models in solving these puzzles, and our assumption of “Minesweeper being a new challenge for GPTs” holds.

C Interaction Examples

In this section, we provide more case studies on the reasoning chains generated by LLMs as support of our discussion in § 5. **For better readability, we convert all coordinate representations to table representations.** Please refer to our [GitHub repository](#) for a complete set of experiment results.

Listing 6: An example of LLM-system interaction with table representation in NC mode from GPT-3.5-16k.

```

1 // Initial Instructions are Omitted
2
3 >> ASSISTANT:
4 REASONING: Since the board is completely covered
  with unopened cells, there is no specific
  information available to make an informed
  decision about which cell to open first.
  Therefore, I will start by randomly selecting a
  cell to open.
5
6 ACTION: L(3,3)
7
8 >> USER:
9 --- CURRENT BOARD ---
10 ---
11 "0","1","2","3","4","5"
12 "1","?","?","?","?","?"

```

```

13 "2","?","?","1","2","?"
14 "3","1","1","1","1","?"
15 "4","1","1","1","1","1"
16 "5","1","1","1","1","1"
17 ---
18
19 please ensure:
20 - You do not duplicate actions.
21 - You submit only one action at a time.
22
23 REASONING:
24
25 ACTION:
26
27
28 >> ASSISTANT:
29 REASONING: Looking at the newly revealed cells, we
  can see that the cell at (2,2) is surrounded by
  two mines. The cell at (2,3) is surrounded by
  one mine. The cell at (3,2) is also surrounded
  by one mine. Based on this information, we can
  deduce that the cell at (3,3) is safe to open.
30
31 ACTION: L(3,3)
32
33 >> USER:
34 Your previous action "L(3,3)" is invalid. Error
  Message:
35 Invalid action: Cannot left-click a blank cell. Left
  -click is only for unopened cells ('?').
36 Please follow the instructions and try again.
37
38 --- CURRENT BOARD ---
39 ---
40 "0","1","2","3","4","5"
41 "1","?","?","?","?","?"
42 "2","?","2","1","2","?"
43 "3","1","1","1","1","?"
44 "4","1","1","1","1","1"
45 "5","1","1","1","1","1"
46 ---
47
48 please ensure:
49 - You do not duplicate actions.
50 - You submit only one action at a time.
51
52 REASONING:
53
54 ACTION:
55
56
57 >> ASSISTANT:
58 REASONING: Apologies for the mistake. Since the cell
  at (3,3) is already opened and blank, we need
  to look for another cell to open. Based on the
  current information, the cell at (4,4) is
  surrounded by one mine and the cell at (4,5) is
  surrounded by one mine. However, the cell at
  (5,5) is surrounded by two mines. Therefore, it
  is safer to open the cell at (5,5) next.
59
60 ACTION: L(5,5)

```

```

61
62 >> USER:
63 Your previous action "L(5,5)" is invalid. Error
  Message:
64 Invalid action: Cannot left-click a blank cell. Left
  -click is only for unopened cells ('?').
65 Please follow the instructions and try again.
66
67 --- CURRENT BOARD ---
68 ---
69 "0","1","2","3","4","5"
70 "1","?","?","?","?","?"
71 "2","?","2","1","2","?"
72 "3","1","1",".","1","?"
73 "4",".",".",".","1","1"
74 "5",".",".",".",".","."
75 ---
76
77 please ensure:
78 - You do not duplicate actions.
79 - You submit only one action at a time.
80
81 REASONING:
82
83 ACTION:
84
85
86 >> ASSISTANT:
87 REASONING: Apologies for the mistake again. Since
  the cell at (5,5) is already opened and blank,
  we need to look for another cell to open. Based
  on the current information, the cell at (4,4)
  is surrounded by one mine and the cell at (4,5)
  is surrounded by one mine. However, the cell
  at (5,4) is surrounded by two mines. Therefore,
  it is safer to open the cell at (5,4) next.
88
89 ACTION: L(5,4)
90
91 >> USER:
92 Your previous action "L(5,4)" is invalid. Error
  Message:
93 Invalid action: Cannot left-click a blank cell. Left
  -click is only for unopened cells ('?').
94 Please follow the instructions and try again.
95
96 --- CURRENT BOARD ---
97 ---
98 "0","1","2","3","4","5"
99 "1","?","?","?","?","?"
100 "2","?","2","1","2","?"
101 "3","1","1",".","1","?"
102 "4",".",".",".","1","1"
103 "5",".",".",".",".","."
104 ---
105
106 please ensure:
107 - You do not duplicate actions.
108 - You submit only one action at a time.
109
110 REASONING:
111
112 ACTION:
113
114
115 >> ASSISTANT:
116 REASONING: Apologies for the mistake once again.
  Since the cell at (5,4) is already opened and
  blank, we need to look for another cell to open
  . Based on the current information, the cell at
  (4,4) is surrounded by one mine and the cell
  at (4,5) is surrounded by one mine. However,
  the cell at (5,3) is surrounded by two mines.
  Therefore, it is safer to open the cell at
  (5,3) next.
117
118 ACTION: L(5,3)
119
120 >> USER:
121 Your previous action "L(5,3)" is invalid. Error
  Message:
122 Invalid action: Cannot left-click a blank cell. Left
  -click is only for unopened cells ('?').
123 Please follow the instructions and try again.
124
125 --- CURRENT BOARD ---
126 ---
127 "0","1","2","3","4","5"
128 "1","?","?","?","?","?"
129 "2","?","2","1","2","?"
130 "3","1","1",".","1","?"
131 "4",".",".",".","1","1"
132 "5",".",".",".",".","."
133 ---
134
135 please ensure:
136 - You do not duplicate actions.
137 - You submit only one action at a time.
138
139 REASONING:
140
141 ACTION:
142
143
144 >> ASSISTANT:
145 REASONING: Apologies for the repeated mistakes.
  Since the cell at (5,3) is already opened and
  blank, we need to look for another cell to open
  . Based on the current information, the cell at
  (4,4) is surrounded by one mine and the cell
  at (4,5) is surrounded by one mine. However,
  the cell at (5,2) is surrounded by two mines.
  Therefore, it is safer to open the cell at
  (5,2) next.
146
147 ACTION: L(5,2)
148
149 >> USER:
150 Your previous action "L(5,2)" is invalid. Error
  Message:
151 Invalid action: Cannot left-click a blank cell. Left
  -click is only for unopened cells ('?').
152 Please follow the instructions and try again.
153
154 --- CURRENT BOARD ---
155 ---
156 "0","1","2","3","4","5"
157 "1","?","?","?","?","?"
158 "2","?","2","1","2","?"
159 "3","1","1",".","1","?"
160 "4",".",".",".","1","1"
161 "5",".",".",".",".","."
162 ---
163
164 please ensure:
165 - You do not duplicate actions.
166 - You submit only one action at a time.
167
168 REASONING:
169
170 ACTION:
171
172
173 >> ASSISTANT:
174 REASONING: Apologies for the repeated mistakes.
  Since the cell at (5,2) is already opened and
  blank, we need to look for another cell to open
  . Based on the current information, the cell at
  (4,4) is surrounded by one mine and the cell
  at (4,5) is surrounded by one mine. However,
  the cell at (5,1) is surrounded by two mines.
  Therefore, it is safer to open the cell at
  (5,1) next.
175
176 ACTION: L(5,1)
177
178 >> USER:
179 Your previous action "L(5,1)" is invalid. Error
  Message:
180 Invalid action: Cannot left-click a blank cell. Left
  -click is only for unopened cells ('?').
181 Please follow the instructions and try again.
182
183 --- CURRENT BOARD ---
184 ---
185 "0","1","2","3","4","5"
186 "1","?","?","?","?","?"
187 "2","?","2","1","2","?"
188 "3","1","1",".","1","?"
189 "4",".",".",".","1","1"
190 "5",".",".",".",".","."
191 ---
192
193 please ensure:

```



```

194 - You do not duplicate actions.
195 - You submit only one action at a time.
196
197 REASONING:
198
199 ACTION:
200
201
202 >> ASSISTANT:
203 REASONING: Apologies for the repeated mistakes.
      Since the cell at (5,1) is already opened and
      blank, we need to look for another cell to open
      . Based on the current information, the cell at
      (4,4) is surrounded by one mine and the cell
      at (4,5) is surrounded by one mine. However,
      the cell at (5,0) is surrounded by two mines.
      Therefore, it is safer to open the cell at
      (5,0) next.
204
205 ACTION: L(5,0)
206
207 >> USER:
208 Your previous action "L(5,0)" is invalid. Error
      Message:
209 Invalid Coordinates! Please make sure your
      coordinate are within [1, 5] for rows and [1,
      5] for columns.
210 Please follow the instructions and try again.
211
212 --- CURRENT BOARD ---
213 ---
214 "0","1","2","3","4","5"
215 "1","?","?","?","?","?"
216 "2","?","2","1","2","?"
217 "3","1","1","?","1","?"
218 "4","?","?","?","1","1"
219 "5","?","?","?","?","?"
220 ---
221
222 please ensure:
223 - You do not duplicate actions.
224 - You submit only one action at a time.
225
226 REASONING:
227
228 ACTION:
229
230
231 >> ASSISTANT:
232 REASONING: Apologies for the repeated mistakes.
      Since the cell at (5,0) is not a valid
      coordinate, we need to look for another cell to
      open. Based on the current information, the
      cell at (4,4) is surrounded by one mine and the
      cell at (4,5) is surrounded by one mine.
      However, the cell at (5,1) is surrounded by two
      mines. Therefore, it is safer to open the cell
      at (5,1) next.
233
234 ACTION: L(5,1)
235
236 >> USER:
237 Your previous action "L(5,1)" is invalid. Error
      Message:
238 Invalid action: Cannot left-click a blank cell. Left
      -click is only for unopened cells ('?').
239 Please follow the instructions and try again.
240
241 --- CURRENT BOARD ---
242 ---
243 "0","1","2","3","4","5"
244 "1","?","?","?","?","?"
245 "2","?","2","1","2","?"
246 "3","1","1","?","1","?"
247 "4","?","?","?","1","1"
248 "5","?","?","?","?","?"
249 ---
250
251 please ensure:
252 - You do not duplicate actions.
253 - You submit only one action at a time.
254
255 REASONING:
256
257 ACTION:
258
259

```

```

260 >> ASSISTANT:
261 REASONING: Apologies for the repeated mistakes.
      Since the cell at (5,1) is already opened and
      blank, we need to look for another cell to open
      . Based on the current information, the cell at
      (4,4) is surrounded by one mine and the cell
      at (4,5) is surrounded by one mine. However,
      the cell at (5,2) is surrounded by two mines.
      Therefore, it is safer to open the cell at
      (5,2) next.
262
263 ACTION: L(5,2)

```

Listing 7: An example of LLM-system interaction with coordinate representation in CH mode from GPT-3.5-instruct.

```

1 // Initial Instruction Duplicates Listing 5 and thus
  Omitted
2
3 --- YOUR ACTION HISTORY ---
4 [Action 1]: ACTION: R(2,2) -> FEEDBACK: Please begin
  by left-clicking on a cell.
5 [Action 2]: ACTION: L(1,1) -> FEEDBACK: Action
  successful!
6 [Action 3]: ACTION: R(5,4) -> FEEDBACK: Action
  successful!
7 [Action 4]: ACTION: R(4,5) -> FEEDBACK: Action
  successful!
8 [Action 5]: ACTION: R(5,5) -> FEEDBACK: Action
  successful!
9 [Action 6]: ACTION: R(5,2) -> FEEDBACK: Action
  successful!
10 [Action 7]: ACTION: R(5,1) -> FEEDBACK: Action
  successful!
11 [Action 8]: ACTION: R(5,5) -> FEEDBACK: Action
  successful!
12 [Action 9]: ACTION: R(5,5) -> FEEDBACK: Action
  successful!
13
14
15 --- CURRENT BOARD ---
16 ---
17 "0","1","2","3","4","5"
18 "1","?","?","?","?","?"
19 "2","?","2","1","2","?"
20 "3","?","?","?","1","1"
21 "4","1","2","2","3","F"
22 "5","F","F","?","F","F"
23 ---
24
25 --- RESPONSE GUIDE ---
26 Let's think step by step.
27
28 Begin by detailing your rationale in the "REASONING"
  section, followed by specifying your move
  using the previously mentioned format in the "
  ACTION" section. Ensure your reasoning doesn't
  exceed 200 words.
29
30 REASONING:
31 ACTION:
32
33
34 >> ASSISTANT:
35
36 REASONING: The cell at (4,5) is a numbered cell with
  a value of "3". This indicates that there are
  three mines in its neighboring cells. Out of
  the three neighboring cells, two are already
  flagged with "F". This means that the remaining
  unopened cell at (5,3) must contain the third
  mine. Therefore, we can safely flag the cell at
  (5,3) with a right-click.
37
38 ACTION: R(5,3)

```

Listing 8: An example of LLM-system interaction with coordinate representation in NC mode from GPT-4.

```

1 // Initial Instructions and Responses are Omitted
2
3 >> USER:
4 --- CURRENT BOARD ---

```

```

5  ---
6  "0","1","2","3","4","5"
7  "1","?",~1',~. ',~1',~?'
8  "2","?",~2',~. ',~1',~?'
9  "3","?",~2',~. ',~1',~?'
10 "4","?",~1',~1',~1',~?'
11 "5","?",~?',~?',~?',~?'
12 ---
13
14 please ensure:
15 - You do not duplicate actions.
16 - You submit only one action at a time.
17
18 REASONING:
19 ACTION:
20
21 >> ASSISTANT:
22 REASONING:
23 The cell at (2,2) indicates there are 2 mines in its
   neighboring cells. Since (1,1), (2,1), and
   (3,1) are the only unopened cells around it, we
   can deduce that two of these cells contain
   mines. Similarly, the cell at (3,2) also
   indicates there are 2 mines in its neighboring
   cells. Considering that (3,1), (4,1), and (5,1)
   are the only unopened cells around it, and we
   know that two of (1,1), (2,1), and (3,1)
   contain mines, we can conclude that (4,1) and
   (5,1) are safe to open.
24
25 ACTION: L(4,1)
26
27 >> USER:
28 --- CURRENT BOARD ---
29 ---
30 "0","1","2","3","4","5"
31 "1","?",~1',~. ',~1',~?'
32 "2","?",~2',~. ',~1',~?'
33 "3","?",~2',~. ',~1',~?'
34 "4",~1',~1',~1',~1',~?'
35 "5","?",~?',~?',~?',~?'
36 ---
37
38 please ensure:
39 - You do not duplicate actions.
40 - You submit only one action at a time.
41
42 REASONING:
43 ACTION:
44
45 >> ASSISTANT:
46 REASONING:
47 The cell at (4,1) indicates there's 1 mine in its
   neighboring cells. Since (3,1), (5,1), and
   (5,2) are the only unopened cells around it,
   and we know that one of (3,1) or (5,1) contains
   a mine (from the previous reasoning), we can
   conclude that (5,2) is safe to open.
48
49 ACTION: L(5,2)
50
51 >> USER:
52 --- CURRENT BOARD ---
53 ---
54 "0","1","2","3","4","5"
55 "1","?",~1',~. ',~1',~?'
56 "2","?",~2',~. ',~1',~?'
57 "3","?",~2',~. ',~1',~?'
58 "4",~1',~1',~1',~1',~?'
59 "5",~. ',~. ',~1',~?',~?'
60 ---
61
62 please ensure:
63 - You do not duplicate actions.
64 - You submit only one action at a time.
65
66 REASONING:
67 ACTION:
68
69 >> ASSISTANT:
70 REASONING:
71 The cell at (5,3) indicates there's 1 mine in its
   neighboring cells. Since (4,4), (5,4), and
   (5,5) are the only unopened cells around it, we
   can deduce that one of these cells contains a
   mine. However, we don't have enough information
   to determine which one. Therefore, the best

```

```

   strategy is to open a cell that is not adjacent
   to any numbered cells. The cell at (1,5) fits
   this criterion.
72
73 ACTION: L(1,5)
74
75 >> USER:
76 --- CURRENT BOARD ---
77 ---
78 "0","1","2","3","4","5"
79 "1","?",~1',~. ',~1',~1'
80 "2","?",~2',~. ',~1',~?'
81 "3","?",~2',~. ',~1',~?'
82 "4",~1',~1',~1',~1',~?'
83 "5",~. ',~. ',~1',~?',~?'
84 ---
85
86 please ensure:
87 - You do not duplicate actions.
88 - You submit only one action at a time.
89
90 REASONING:
91 ACTION:
92
93 >> ASSISTANT:
94 REASONING:
95 The cell at (1,5) indicates there's 1 mine in its
   neighboring cells. Since (2,5) is the only
   unopened cell around it, we can deduce that
   (2,5) contains a mine. Therefore, our next move
   is to flag the cell at (2,5) with a right-
   click.
96
97 ACTION: R(2,5)
98
99 >> USER:
100 --- CURRENT BOARD ---
101 ---
102 "0","1","2","3","4","5"
103 "1","?",~1',~. ',~1',~1'
104 "2","?",~2',~. ',~1',~F'
105 "3","?",~2',~. ',~1',~?'
106 "4",~1',~1',~1',~1',~?'
107 "5",~. ',~. ',~1',~?',~?'
108 ---
109
110 >> ASSISTANT:
111 REASONING:
112 The cell at (2,4) indicates there's 1 mine in its
   neighboring cells. Since (2,5) is flagged and
   (3,5) is the only other unopened cell around it
   , we can deduce that (3,5) does not contain a
   mine. Therefore, our next move is to open the
   cell at (3,5) with a left-click.
113
114 ACTION: L(3,5)
115
116 >> USER:
117 --- CURRENT BOARD ---
118 ---
119 "0","1","2","3","4","5"
120 "1","?",~1',~. ',~1',~1'
121 "2","?",~2',~. ',~1',~F'
122 "3","?",~2',~. ',~1',~1'
123 "4",~1',~1',~1',~1',~?'
124 "5",~. ',~. ',~1',~?',~?'
125 ---
126
127 please ensure:
128 - You do not duplicate actions.
129 - You submit only one action at a time.
130
131 REASONING:
132 ACTION:
133
134 >> ASSISTANT:
135 REASONING:
136 The cell at (3,5) indicates there's 1 mine in its
   neighboring cells. Since (4,5) is the only
   unopened cell around it, we can deduce that
   (4,5) contains a mine. Therefore, our next move
   is to flag the cell at (4,5) with a right-
   click.
137
138 ACTION: R(4,5)
139
140 >> USER:

```

```

141 --- CURRENT BOARD ---
142 ---
143 "0","1","2","3","4","5"
144 "1","?",~1',~1',~1',~1',~1'
145 "2","?",~2',~1',~1',~F'
146 "3","?",~2',~1',~1',~1'
147 "4","1',~1',~1',~1',~F'
148 "5",~1',~1',~1',~?',~?'
149 ---
150
151 >> ASSISTANT:
152 REASONING:
153 The cell at (5,3) indicates there's 1 mine in its
    neighboring cells. Since (4,5) is flagged and
    (5,4) and (5,5) are the only other unopened
    cells around it, we can deduce that one of them
    contains a mine. However, we don't have enough
    information to determine which one. Therefore,
    the best strategy is to open a cell that is
    not adjacent to any numbered cells. The cell at
    (1,1) fits this criterion.
154
155 ACTION: L(1,1)
156
157 >> USER:
158 --- CURRENT BOARD ---
159 ---
160 "0","1","2","3","4","5"
161 "1",~1',~1',~1',~1',~1'
162 "2","?",~2',~1',~1',~F'
163 "3","?",~2',~1',~1',~1'
164 "4","1',~1',~1',~1',~F'
165 "5",~1',~1',~1',~?',~?'
166 ---
167
168 >> ASSISTANT:
169 REASONING:
170 The cell at (1,1) indicates there's 1 mine in its
    neighboring cells. Since (2,1) is the only
    unopened cell around it, we can deduce that
    (2,1) contains a mine. Therefore, our next move
    is to flag the cell at (2,1) with a right-
    click.
171
172 ACTION: R(2,1)
173
174 >> USER:
175 --- CURRENT BOARD ---
176 ---
177 "0","1","2","3","4","5"
178 "1",~1',~1',~1',~1',~1'
179 "2",~F',~2',~1',~1',~F'
180 "3","?",~2',~1',~1',~1'
181 "4","1',~1',~1',~1',~F'
182 "5",~1',~1',~1',~?',~?'
183 ---
184
185 please ensure:
186 - You do not duplicate actions.
187 - You submit only one action at a time.
188
189 REASONING:
190 ACTION:
191
192 >> ASSISTANT:
193 REASONING:
194 The cell at (3,2) indicates there are 2 mines in its
    neighboring cells. Since (2,1) and (2,5) are
    flagged and (3,1) is the only other unopened
    cell around it, we can deduce that (3,1) does
    not contain a mine. Therefore, our next move is
    to open the cell at (3,1) with a left-click.
195
196 ACTION: L(3,1)

```